

torch.linalg.solve_ex#

https://pytorch.org/docs/stable/generated/torch.linalg.solve_ex.html

Description:

A version of `solve()` that does not perform error checks unless `check_errors= True`. It also returns the info tensor returned by LAPACK's `getrf`. When the inputs are on a CUDA device, this function synchronizes only when `check_errors= True`. This function is “experimental” and it may change in a future PyTorch release.

Function Signature

```
torch.linalg.solve_ex(A, B, *, left=True,  
check_errors=False, out=None)#
```

Parameters

Keyword Arguments

Returns

Parameters

Keyword

`left` (bool, optional) – whether to solve the system $AX=B$ or $XA=B$. Default: True. `check_errors` (bool, optional) – controls whether to check the content of `infos` and raise an error if it is non-zero. Default: False. `out` (tuple, optional) – tuple of two tensors to write the output to. Ignored if None. Default: None.

Returns:

A named tuple (result, info).

Code Examples

```
>>> A = torch.randn(3, 3)
>>> Ainv, info = torch.linalg.solve_ex(A)
>>> torch.dist(torch.linalg.inv(A), Ainv)
tensor(0.)
>>> info
tensor(0, dtype=torch.int32)
```

Notes & Warnings

NOTE

Note When the inputs are on a CUDA device, this function synchronizes only when `check_errors= True`.

⚠ WARNING

Warning This function is “experimental” and it may change in a future PyTorch release.

Detailed Documentation

Documentation

A version of `solve()` that does not perform error checks unless `check_errors= True`. It also returns the info tensor returned by LAPACK's `getrf`.

When the inputs are on a CUDA device, this function synchronizes only when `check_errors= True`.

This function is “experimental” and it may change in a future PyTorch release.

A (Tensor) – tensor of shape $(*, n, n)$ where $*$ is zero or more batch dimensions.

left (bool, optional) – whether to solve the system $AX=B$ or $XA=B$. Default: True.

check_errors (bool, optional) – controls whether to check the content of infos and raise an error if it is non-zero. Default: False.

out (tuple, optional) – tuple of two tensors to write the output to. Ignored if None. Default: None.

A named tuple (result, info).